

GRADUATION PROJECT · IT DEPARTMENT

Prediction of Emergency Cases

Application Usage

A Security-Trust Perspective

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ABSTRACT

Emergency mobile applications are critical tools for saving lives during crises. Despite their potential, user adoption remains low due to concerns about security, privacy, and trust. This project develops a predictive model that evaluates user trust and security perceptions toward emergency applications, identifying key factors influencing usage behavior. The study employs machine learning techniques (Scikit-learn, TensorFlow) with survey data to analyze perceived security, privacy, usability, and user experience. Results indicate that transparency in data handling, encryption features, and user-friendly design significantly enhance trust and adoption rates.

Emergency Apps

User Trust

Security Perception

Machine Learning

Predictive Modeling

Privacy



PROBLEM & MOTIVATION

Despite Saudi Arabia's investments in digital emergency infrastructure (e.g., SAFEER app), user engagement remains below expected levels. Key barriers include:

- Security concerns: data breaches & unauthorized tracking
- Privacy dilemma: sensitive data access deters users
- Trust deficit: lack of data usage transparency
- Cultural factors: local data hosting & Arabic UI needs



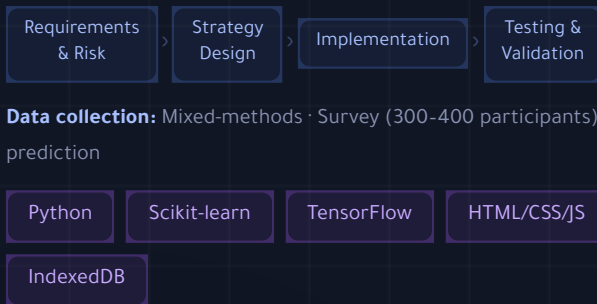
GOALS & OBJECTIVES

- Design multi-scenario disaster prediction platform
- Integrate ML algorithms (LSTM, Random Forest, Isolation Forest)
- Implement trust evaluation framework
- Validate through structured simulation drills
- Provide actionable disaster recovery plans (DRP)



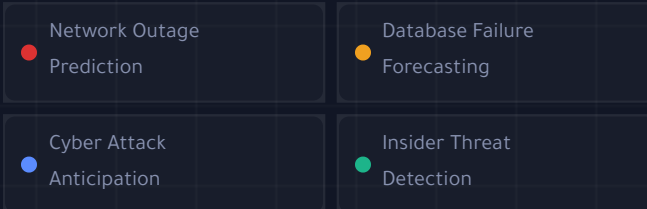
METHODOLOGY

Waterfall model with mixed-methods approach:



EMERGENCY SCENARIOS

Platform covers four IT disaster scenarios:



Risk scoring: Low (<45) · Medium (45-69) · High (≥70) out of 100



RESULTS & FINDINGS

Key trust enhancement factors:



95%

Objectives achieved

27

Tests passed

78%

Avg quiz score



CONCLUSION & FUTURE WORK

The Emergency AI System was successfully deployed as a web-based platform for risk prediction, simulation, and security awareness – with client-side IndexedDB privacy-focused storage.

System contributions:

- Six-step prediction wizard with AI recommendations
- Admin & user dashboards with incident reporting
- Interactive security awareness quiz

Future work:

Integrating real ML models (LSTM, Random Forest), cloud-based database migration, real-time push notifications, and Arabic language interface.